Band-1 Hadamard order 14564 – Classical Paley-I (q=14563); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14564 Paley-I PASS; Paley-II FAIL/SKIP (q=7281=3^2*809 NOT prime power; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4 3641; 3641===1 mod 4 odd NOT Hadamard-admissible; 3641=11*331); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-13

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. 2026-10-13_Band1_Hadamard_14552_PaleyI; 2026-10-13_Band1_Hadamard_14544_PaleyI; 2026-10-12_Band1_Hadamard_14520_PaleyI; 2026-10-12_Band1_Hadamard_14508_PaleyII; 2026-10-12_Band1_Hadamard_14504_PaleyI; 2026-10-12_Band1_Hadamard_14480_PaleyI; 2026-10-12_Band1_Hadamard_14476_PaleyII; 2026-10-09_Band1_Hadamard_14460_PaleyII; 2026-10-09_Band1_Hadamard_14448_PaleyI; 2026-10-09_Band1_Hadamard_14432_PaleyI; 2026-10-09_Band1_Hadamard_14428_PaleyII; 2026-10-09_Band1_Hadamard_14424_PaleyI; 2026-10-09_Band1_Hadamard_14420_PaleyI; 2026-10-09_Band1_Hadamard_14412_PaleyI; 2026-10-09_Band1_Hadamard_14408_PaleyI; 2026-10-09_Band1_Hadamard_14388_PaleyI; 2026-10-09_Band1_Hadamard_14356_PaleyII; 2026-10-09_Band1_Hadamard_14348_PaleyI; 2026-10-09_Band1_Hadamard_14328_PaleyI; 2026-10-09_Band1_Hadamard_14324_PaleyI; 2026-10-09_Band1_Hadamard_14304_PaleyI; 2026-10-09_Band1_Hadamard_14252_PaleyI; 2026-10-09_Band1_Hadamard_14244_PaleyI; 2026-10-09_Band1_Hadamard_14208_PaleyI; 2026-10-08_Band1_Hadamard_14160_PaleyI; 2026-10-08_Band1_Hadamard_14144_PaleyI; 2026-10-08_Band1_Hadamard_14108_PaleyI; 2026-10-08_Band1_Hadamard_14088_PaleyI; 2026-10-08_Band1_Hadamard_14084_PaleyI; 2026-10-08_Band1_Hadamard_14072_PaleyI; 2026-10-08_Band1_Hadamard_14052_PaleyI; 2026-10-08_Band1_Hadamard_14012_PaleyI; 2026-10-07_Band1_Hadamard_14000_PaleyI; 2026-10-07_Band1_Hadamard_13968_PaleyI; 2026-10-07_Band1_Hadamard_13964_PaleyI; 2026-10-07_Band1_Hadamard_13932_PaleyI; 2026-10-07_Band1_Hadamard_13908_PaleyI; 2026-10-07_Band1_Hadamard_13904_PaleyI; 2026-10-07_Band1_Hadamard_13884_PaleyI; 2026-10-07_Band1_Hadamard_13880_PaleyI; 2026-10-07_Band1_Hadamard_13860_PaleyI; 2026-10-07_Band1_Hadamard_13832_PaleyI; 2026-10-07_Band1_Hadamard_13808_PaleyI; 2026-10-07_Band1_Hadamard_13800_PaleyI; 2026-10-07_Band1_Hadamard_13764_PaleyI; 2026-10-07_Band1_Hadamard_13760_PaleyI; 2026-10-07_Band1_Hadamard_13752_PaleyI; 2026-10-07_Band1_Hadamard_13724_PaleyI;

2026-10-07_Band1_Hadamard_13712_PaleyI; 2026-10-07_Band1_Hadamard_13692_PaleyI;
2026-10-07_Band1_Hadamard_13688_PaleyI; 2026-10-06_Band1_Hadamard_13680_PaleyI;
2026-10-06_Band1_Hadamard_13628_PaleyI; 2026-10-06_Band1_Hadamard_13620_PaleyI;
2026-10-06_Band1_Hadamard_13592_PaleyI; 2026-10-06_Band1_Hadamard_13568_PaleyI;
2026-10-06_Band1_Hadamard_13524_PaleyI; 2026-10-06_Band1_Hadamard_13500_PaleyI;
2026-10-06_Band1_Hadamard_13488_PaleyI; 2026-10-06_Band1_Hadamard_13464_PaleyI;
2026-10-06_Band1_Hadamard_13452_PaleyI; 2026-10-06_Band1_Hadamard_13412_PaleyI;
2026-10-06_Band1_Hadamard_13400_PaleyI; 2026-10-06_Band1_Hadamard_13368_PaleyI;
2026-10-06_Band1_Hadamard_13340_PaleyI; 2026-10-06_Band1_Hadamard_13332_PaleyI;
2026-10-06_Band1_Hadamard_13328_PaleyI; 2026-10-06_Band1_Hadamard_13292_PaleyI;
2026-10-06_Band1_Hadamard_13268_PaleyI; 2026-10-06_Band1_Hadamard_13260_PaleyI;
2026-10-06_Band1_Hadamard_13220_PaleyI; 2026-10-06_Band1_Hadamard_13188_PaleyI;
2026-10-06_Band1_Hadamard_13184_PaleyI; 2026-10-06_Band1_Hadamard_13172_PaleyI;
2026-10-05_Band1_Hadamard_13164_PaleyI; 2026-10-05_Band1_Hadamard_13160_PaleyI;
2026-10-05_Band1_Hadamard_13152_PaleyI; 2026-10-05_Band1_Hadamard_13148_PaleyI;
2026-10-05_Band1_Hadamard_13128_PaleyI; 2026-10-05_Band1_Hadamard_13104_PaleyI;
2026-10-05_Band1_Hadamard_13100_PaleyI; 2026-10-05_Band1_Hadamard_13064_PaleyI;
2026-10-05_Band1_Hadamard_13044_PaleyI; 2026-10-05_Band1_Hadamard_13008_PaleyI;
2026-10-05_Band1_Hadamard_13004_PaleyI; 2026-10-05_Band1_Hadamard_12984_PaleyI;
2026-10-05_Band1_Hadamard_12980_PaleyI; 2026-10-05_Band1_Hadamard_12968_PaleyI;
2026-10-05_Band1_Hadamard_12960_PaleyI; 2026-10-05_Band1_Hadamard_12924_PaleyI;
2026-10-05_Band1_Hadamard_12920_PaleyI; 2026-10-05_Band1_Hadamard_12912_PaleyI;
2026-10-05_Band1_Hadamard_12908_PaleyI; 2026-10-04_Band1_Hadamard_12900_PaleyI;
2026-10-04_Band1_Hadamard_12824_PaleyI; 2026-10-04_Band1_Hadamard_12800_PaleyI;
2026-10-04_Band1_Hadamard_12792_PaleyI; 2026-10-04_Band1_Hadamard_12764_PaleyI;
2026-10-04_Band1_Hadamard_12744_PaleyI; 2026-10-04_Band1_Hadamard_12740_PaleyI;
2026-10-04_Band1_Hadamard_12704_PaleyI; 2026-10-04_Band1_Hadamard_12672_PaleyI;
2026-10-04_Band1_Hadamard_12660_PaleyI; 2026-10-04_Band1_Hadamard_12648_PaleyI;
2026-10-04_Band1_Hadamard_12620_PaleyI; 2026-10-04_Band1_Hadamard_12612_PaleyI;
2026-10-04_Band1_Hadamard_12584_PaleyI; 2026-10-04_Band1_Hadamard_12548_PaleyI;
2026-10-04_Band1_Hadamard_12540_PaleyI; 2026-10-03_Band1_Hadamard_12380_PaleyI;
2026-10-03_Band1_Hadamard_12348_PaleyI; 2026-10-03_Band1_Hadamard_12344_PaleyI;
2026-10-03_Band1_Hadamard_12324_PaleyI; 2026-10-03_Band1_Hadamard_12264_PaleyI;
2026-10-03_Band1_Hadamard_12252_PaleyI; 2026-10-03_Band1_Hadamard_12240_PaleyI;
2026-10-03_Band1_Hadamard_12228_PaleyI; 2026-10-03_Band1_Hadamard_12212_PaleyI;
2026-10-03_Band1_Hadamard_12204_PaleyI; 2026-10-03_Band1_Hadamard_12164_PaleyI;
2026-10-03_Band1_Hadamard_12144_PaleyI; 2026-10-03_Band1_Hadamard_12120_PaleyI;
2026-10-03_Band1_Hadamard_12108_PaleyI; 2026-10-03_Band1_Hadamard_12072_PaleyI;
2026-10-03_Band1_Hadamard_12044_PaleyI; 2026-10-03_Band1_Hadamard_12012_PaleyI;
2026-10-03_Band1_Hadamard_12008_PaleyI; 2026-10-03_Band1_Hadamard_11988_PaleyI;
2026-10-03_Band1_Hadamard_11972_PaleyI; 2026-10-03_Band1_Hadamard_11960_PaleyI;

2026-10-03_Band1_Hadamard_11940_PaleyI; 2026-10-03_Band1_Hadamard_11928_PaleyI;
2026-10-02_Band1_Hadamard_11924_PaleyI; 2026-10-02_Band1_Hadamard_11904_PaleyI;
2026-10-02_Band1_Hadamard_11888_PaleyI; 2026-10-02_Band1_Hadamard_11868_PaleyI;
2026-10-02_Band1_Hadamard_11864_PaleyI; 2026-10-02_Band1_Hadamard_11840_PaleyI;
2026-10-02_Band1_Hadamard_11832_PaleyI; 2026-10-02_Band1_Hadamard_11828_PaleyI;
2026-10-02_Band1_Hadamard_11808_PaleyI; 2026-10-02_Band1_Hadamard_11784_PaleyI;
2026-10-02_Band1_Hadamard_11780_PaleyI; 2026-10-02_Band1_Hadamard_11744_PaleyI;
2026-10-02_Band1_Hadamard_11732_PaleyI; 2026-10-02_Band1_Hadamard_11720_PaleyI;
2026-10-02_Band1_Hadamard_11700_PaleyI; 2026-09-30_Band1_Hadamard_11588_PaleyI;
2026-09-30_Band1_Hadamard_11580_PaleyI; 2026-09-30_Band1_Hadamard_11552_PaleyI;
2026-09-30_Band1_Hadamard_11528_PaleyI; 2026-09-30_Band1_Hadamard_11520_PaleyI;
2026-09-30_Band1_Hadamard_11504_PaleyI; 2026-09-30_Band1_Hadamard_11492_PaleyI;
2026-09-30_Band1_Hadamard_11484_PaleyI; 2026-09-30_Band1_Hadamard_11472_PaleyI;
2026-09-30_Band1_Hadamard_11468_PaleyI; 2026-09-30_Band1_Hadamard_11448_PaleyI;
2026-09-30_Band1_Hadamard_11444_PaleyI; 2026-09-30_Band1_Hadamard_11424_PaleyI;
2026-09-30_Band1_Hadamard_11412_PaleyI; 2026-09-30_Band1_Hadamard_11400_PaleyI;
2026-09-30_Band1_Hadamard_11384_PaleyI; 2026-09-30_Band1_Hadamard_11352_PaleyI;
2026-09-30_Band1_Hadamard_11312_PaleyI; 2026-09-30_Band1_Hadamard_11300_PaleyI;
2026-09-30_Band1_Hadamard_11288_PaleyI; 2026-09-30_Band1_Hadamard_11280_PaleyI;
2026-09-30_Band1_Hadamard_11252_PaleyI; 2026-09-29_Band1_Hadamard_11244_PaleyI;
2026-09-29_Band1_Hadamard_11240_PaleyI; 2026-09-29_Band1_Hadamard_11172_PaleyI;
2026-09-29_Band1_Hadamard_11160_PaleyI; 2026-09-29_Band1_Hadamard_11132_PaleyI;
2026-09-29_Band1_Hadamard_11120_PaleyI; 2026-09-29_Band1_Hadamard_11088_PaleyI;
2026-09-29_Band1_Hadamard_11084_PaleyI; 2026-09-29_Band1_Hadamard_11072_PaleyI;
2026-09-29_Band1_Hadamard_11060_PaleyI; 2026-09-29_Band1_Hadamard_11048_PaleyI;
2026-09-29_Band1_Hadamard_11028_PaleyI; 2026-09-29_Band1_Hadamard_11004_PaleyI;
2026-09-29_Band1_Hadamard_10988_PaleyI; 2026-09-29_Band1_Hadamard_10980_PaleyI;
2026-09-29_Band1_Hadamard_10940_PaleyI; 2026-09-29_Band1_Hadamard_10892_PaleyI;
2026-09-29_Band1_Hadamard_10884_PaleyI; 2026-09-29_Band1_Hadamard_10860_PaleyI;
2026-09-29_Band1_Hadamard_10848_PaleyI; 2026-09-29_Band1_Hadamard_10832_PaleyI;
2026-09-29_Band1_Hadamard_10800_PaleyI; 2026-09-29_Band1_Hadamard_10772_PaleyI;
2026-09-29_Band1_Hadamard_10740_PaleyI; 2026-09-29_Band1_Hadamard_10724_PaleyI;
2026-09-29_Band1_Hadamard_10712_PaleyI; 2026-09-28_Band1_Hadamard_10692_PaleyI;
2026-09-28_Band1_Hadamard_10688_PaleyI; 2026-09-28_Band1_Hadamard_10668_PaleyI;
2026-09-28_Band1_Hadamard_10652_PaleyI; 2026-09-28_Band1_Hadamard_10640_PaleyI;
2026-09-28_Band1_Hadamard_10632_PaleyI; 2026-09-28_Band1_Hadamard_10628_PaleyI;
2026-09-28_Band1_Hadamard_10608_PaleyI; 2026-09-28_Band1_Hadamard_10568_PaleyI;
2026-09-28_Band1_Hadamard_10560_PaleyI; 2026-09-28_Band1_Hadamard_10532_PaleyI;
2026-09-28_Band1_Hadamard_10500_PaleyI; 2026-09-28_Band1_Hadamard_10488_PaleyI;
2026-09-28_Band1_Hadamard_10464_PaleyI; 2026-09-28_Band1_Hadamard_10460_PaleyI;
2026-09-28_Band1_Hadamard_10428_PaleyI; 2026-09-28_Band1_Hadamard_10400_PaleyI;

2026-09-24_Band1_Hadamard_10392_PaleyI; 2026-09-24_Band1_Hadamard_10344_PaleyI;
2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
2026-09-23_Band1_Hadamard_10164_PaleyI; 2026-09-23_Band1_Hadamard_10160_PaleyI;
2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;
2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI;
2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.
Lab. /workspace/experiments/band1/hadamard_14564_paleyI/.
Cutoff. 2026-10-13T11:34:13Z (~7:34AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14564 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14563 (prime ===3 mod 4). 14564 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14564: Paley-I **PASS** ($q = 14563$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7281=3ˆ2*809 NOT a prime power; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** ($14564 = 4 \cdot 3641$; $3641 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3641 = 11 \cdot 331$; do NOT invent H_3641). **14564** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture; 14564 is a long-known Hadamard order and this is a finite on-box verification, NOT a new-order claim. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14552+14544+14520+14508+14504+14480+14476+14460+14448+14432+14428+14424+14420+144** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14564**.

- H_{14564} = Paley-I ($q = 14563$) – classical SPK
- Paley-II for 14564: **FAIL/SKIP** ($q = 7281 = 3^2 \cdot 809$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 14564: **FAIL/SKIP** (4·3641; $3641 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3641 = 11 \cdot 331$)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_14564.py
```

Results. overall_pass=True; units 820/820; row blocks 114/114 (block size 128); wall 2487.97s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-13T11:34:13Z (~7:34AM ET)

Arithmetic

- Target $n = 14564 = q + 1$ with $q = 14563$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7281 = 3^2 \cdot 809$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $14564 = 4 \cdot 3641$; $3641 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3641 = 11 \cdot 331$ — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14564 Paley-I	Paley-I q=14563	0	2449.0083
H_14564 Paley-II	FAIL/SKIP (q=7281=3ˆ2*809 NOT prime power; ===1 mod 4 but composite)	–	–
H_14564 Kronecker	FAIL/SKIP (4*3641; 3641===1 mod 4 odd NOT Hadamard-admissible; 3641=11*331)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**;

9324; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
14564 in SPK	**True** (Paley-I q=14563)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)
- 14564 is a long-known Hadamard order (Paley 1933 family); finite on-box verification only, NOT a new-order claim

Next optional

NEXT direct route after 14564 (FIRST any direct = Paley-I): q=14591->n=14592 Paley-I DIRECT (prime ===3 mod 4; Paley-II q2=7295=5*1459 composite -> FAIL/SKIP; Kronecker 14592=4*3648; check separately; template hadamard_14564_paleyI / hadamard_14420_paleyI) | first Paley-II after 14564 = q2=7297->n=14596 | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; note-only) | skipped Paley-II DIRECT orders below 14564 still unbuilt (template hadamard_14508_paleyII / hadamard_14476_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)